# Methods results and limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

## One complete methods/results/limitations annotation record.

Keep supplied statement, role, reason, denominator, faithful rewrite and untested explanation together.

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**Keep one habit:** Recall → W1–W3/W5 → independent cutoff/reviewer case → report-reading project → delayed return.

MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

### Finding primary research

> **My remembered idea, example and correction**

### A question that evidence can answer

> **My remembered idea, example and correction**

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MAIN LESSON · SECTION 5

## W1 · Explain the current example

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Methods describe what was done, results describe reported observations with metric/denominator, limitations restrict interpretation, and mechanism hypotheses are explanations still to test. Main supplied passage: 20 fictional queries is method/scope; 16 passed gives 16/20 = 80% result; real student use not studied limits classroom-learning claims.

  Full rubric/raw case identities are not supplied by that short passage. Separate picture evaluates two frozen classifiers on the same five labelled leaf photos from one camera, no outdoor photos. A 3/5=60%, B 4/5=80%, difference 20 percentage points, is a fixed-set result. Missing outdoor/other-camera evidence prevents an outdoor-best or stable-general-ranking claim, not proof B fails outside.

  One statement may serve more than one role: a sampling or reviewer detail is method and may explain a limitation. A single reviewer does not prove wrong labels; agreement remains unassessed unless checked. Complete annotation/rewrite investigation uses only supplied passages and paper arithmetic; no model, participant, causal-mechanism or reviewer-agreement study is performed.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

MAIN LESSON · SECTION 6

# W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

## Complete task and evidence

- Methods describe what was done, results describe reported observations with metric/denominator, limitations restrict interpretation, and mechanism hypotheses are explanations still to test. Main supplied passage: 20 fictional queries is method/scope; 16 passed gives 16/20 = 80% result; real student use not studied limits classroom-learning claims.

  Full rubric/raw case identities are not supplied by that short passage. Separate picture evaluates two frozen classifiers on the same five labelled leaf photos from one camera, no outdoor photos. A 3/5=60%, B 4/5=80%, difference 20 percentage points, is a fixed-set result. Missing outdoor/other-camera evidence prevents an outdoor-best or stable-general-ranking claim, not proof B fails outside.

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- Statements: used 20 fictional queries; 16 passed; real student use was not studied. Show the intermediate steps and give the conclusion with its scope.

**My reasoning and evidence**

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MAIN LESSON · SECTION 7

# W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Methods describe what was done, results describe reported observations with metric/denominator, limitations restrict interpretation, and mechanism hypotheses are explanations still to test. Main supplied passage: 20 fictional queries is method/scope; 16 passed gives 16/20 = 80% result; real student use not studied limits classroom-learning claims.

  Full rubric/raw case identities are not supplied by that short passage. Separate picture evaluates two frozen classifiers on the same five labelled leaf photos from one camera, no outdoor photos. A 3/5=60%, B 4/5=80%, difference 20 percentage points, is a fixed-set result. Missing outdoor/other-camera evidence prevents an outdoor-best or stable-general-ranking claim, not proof B fails outside.

  One statement may serve more than one role: a sampling or reviewer detail is method and may explain a limitation. A single reviewer does not prove wrong labels; agreement remains unassessed unless checked. Complete annotation/rewrite investigation uses only supplied passages and paper arithmetic; no model, participant, causal-mechanism or reviewer-agreement study is performed.

- A report omits this qualification: A **limitation** should constrain a claim rather than be hidden in unrelated prose. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**My reasoning and evidence**

MAIN LESSON · SECTION 9

# W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Methods describe what was done, results describe reported observations with metric/denominator, limitations restrict interpretation, and mechanism hypotheses are explanations still to test. Main supplied passage: 20 fictional queries is method/scope; 16 passed gives 16/20 = 80% result; real student use not studied limits classroom-learning claims.

  Full rubric/raw case identities are not supplied by that short passage. Separate picture evaluates two frozen classifiers on the same five labelled leaf photos from one camera, no outdoor photos. A 3/5=60%, B 4/5=80%, difference 20 percentage points, is a fixed-set result. Missing outdoor/other-camera evidence prevents an outdoor-best or stable-general-ranking claim, not proof B fails outside.

  One statement may serve more than one role: a sampling or reviewer detail is method and may explain a limitation. A single reviewer does not prove wrong labels; agreement remains unassessed unless checked. Complete annotation/rewrite investigation uses only supplied passages and paper arithmetic; no model, participant, causal-mechanism or reviewer-agreement study is performed.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

### My reasoning and evidence

## Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

### Complete task and evidence

- Statements: fixed cutoff 5; 12/15 relevant returned items; judgments came from one reviewer. Classify.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

#### Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

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MAIN LESSON · SECTION 10

## Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

## Complete task and evidence

- Add a labelled **method**s **result**s and **limitation**s **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **method**s **result**s and **limitation**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and **limitation**. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

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MAIN LESSON · SECTION 11

# Feedback · Repair and return

> Keep this task's inputs,rule and response together.

## Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |

| Evidence | My record |
| --- | --- |
| What I can now explain without notes | |
| One remaining question | |

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Use W2: Trace the worked case for **Method**s **result**s and **limitation**s. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

## Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Use the worked case for **Method**s **result**s and **limitation**s. Proposed explanation: "The worked **result** establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

## Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

### Complete task and evidence

- Statements: fixed cutoff 5; 12/15 relevant returned items; judgments came from one reviewer. Classify.

**My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

## Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

### Complete task and evidence

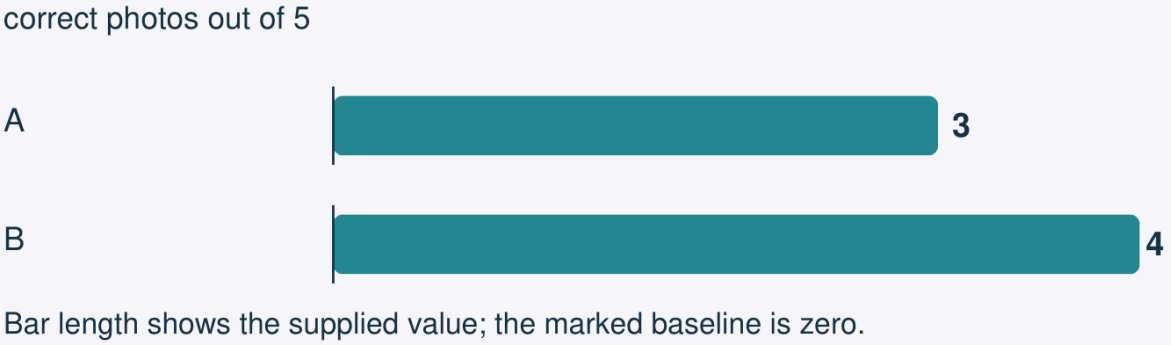
- Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.
- A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.
- Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the **result** establish that B is best for outdoor farming?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

AI ACADEMY / AI-7/4

## Two models on five shared photos




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/4

# Methods results and limitations

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Identify the method

Same five photos, labels and scoring are used for both models.



## 2. State the result

A scores 60%; B scores 80% on this fixed set.



## 3. State the limit

The set supplies no outdoor-camera performance evidence.

**CHANGE ONE THING**

The conclusion says B is best for outdoor farming.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

REFERENCE · SECTION 13

# Reference · Project milestone

> Keep this task’s inputs,rule and response together.

## Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting **method**.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

| Evidence to keep | My record |
|---|---|
| This week’s input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

# Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

## First supported question and optional hint

- Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.
- Which labelled step matches this detail? The set supplies no outdoor-camera performance **evidence**.
- 1. State the **result**
- 2. State the limit
- 3. Identify the **method**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: State the limit The set supplies no outdoor-camera performance **evidence**.

**Method**s tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

## Changed-case question and optional hint

- Changed-case practice: The conclusion says B is best for outdoor farming. Which conclusion is supported by the supplied facts?
- 1. Reject that extension without outdoor **evidence**.
- 2. This one example guarantees correct **result**s for every future **input**.
- 3. We can decide the **result** without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Reject that extension without outdoor **evidence**. The reported conditions do not include the proposed use population.

## Repair and retain the record

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which **method**s and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

## Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

## 01. Methods results and limitations: the case

AI ACADEMY / AI-7/4 · 01 · LOOK AT THE FACTS

### Methods results and limitations: the case

**The situation**

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.

**Supplied evidence**

A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.

**Explain methods results and limitations using the records and boundaries of this fictional investigation.**

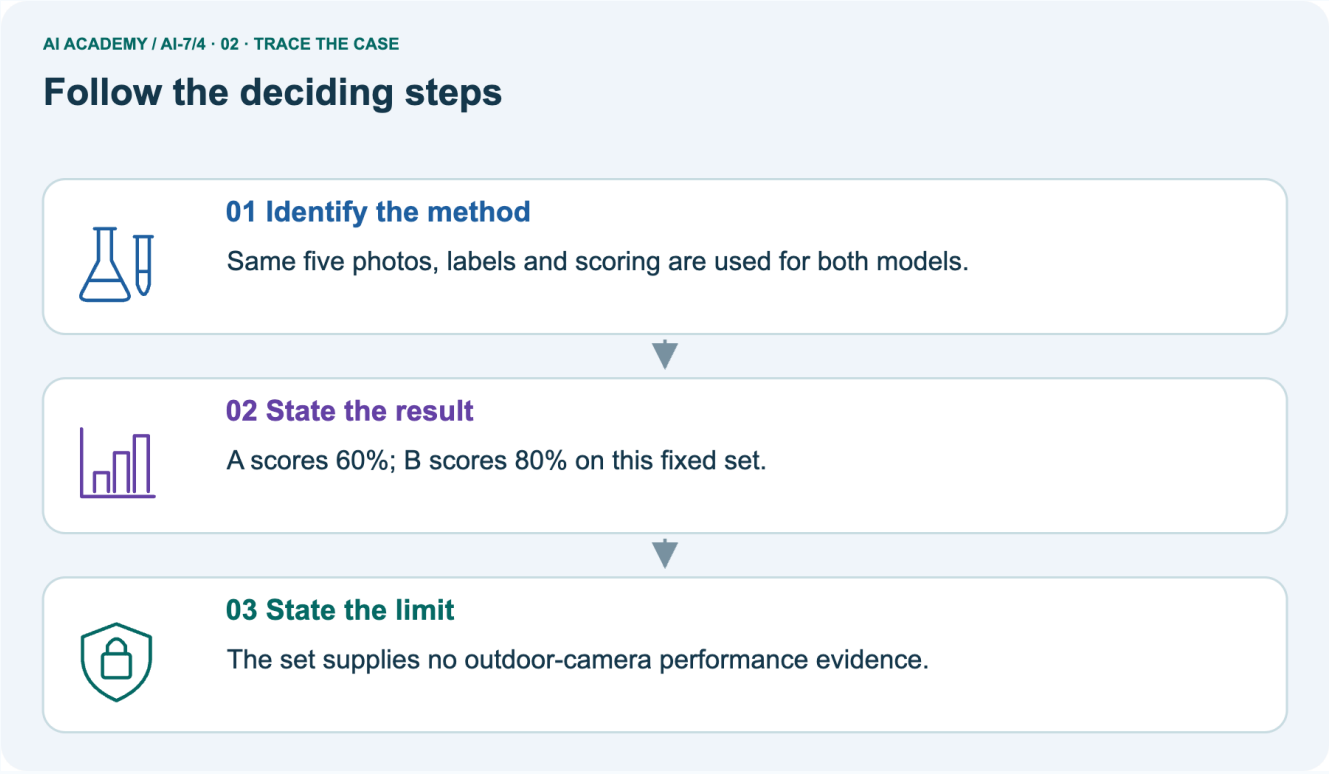
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera. Supplied facts: A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.

Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



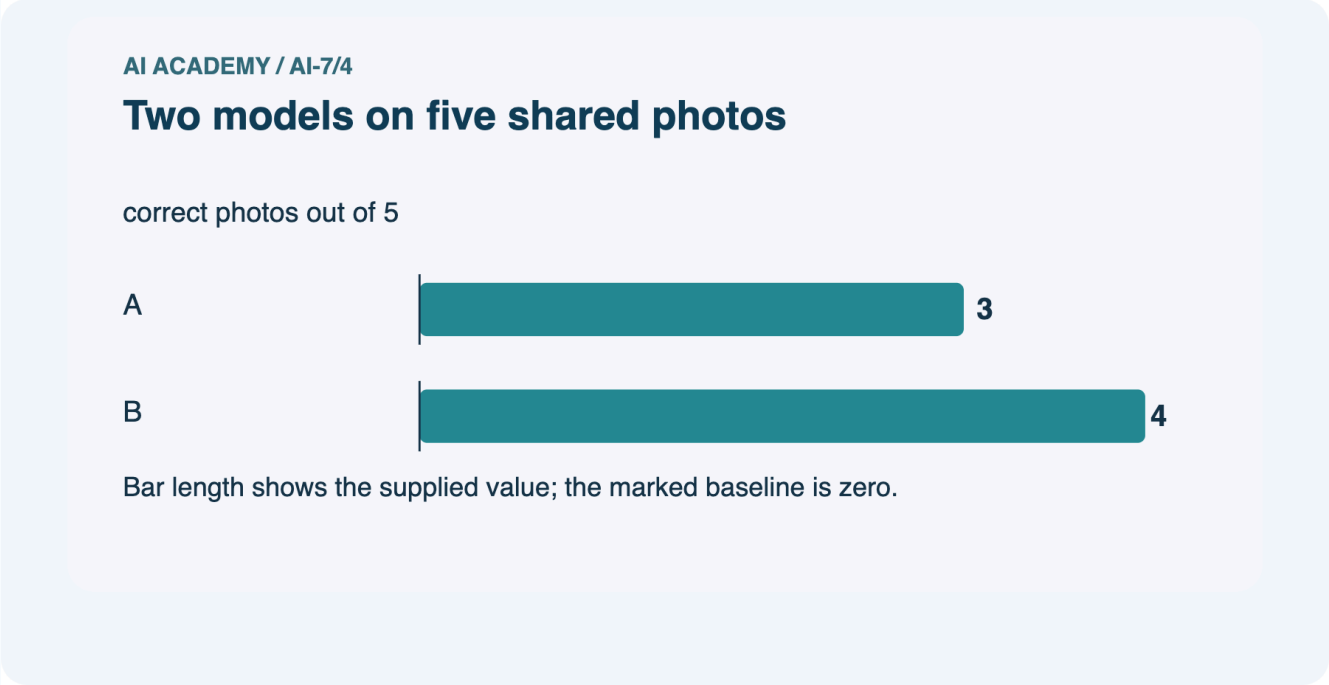

Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the **method**: Same five photos, labels and scoring are used for both models. → State the **result**: A scores **60%**; B scores **80%** on this fixed set. → State the limit: The set supplies no outdoor-camera performance **evidence**.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Two models on five shared photos



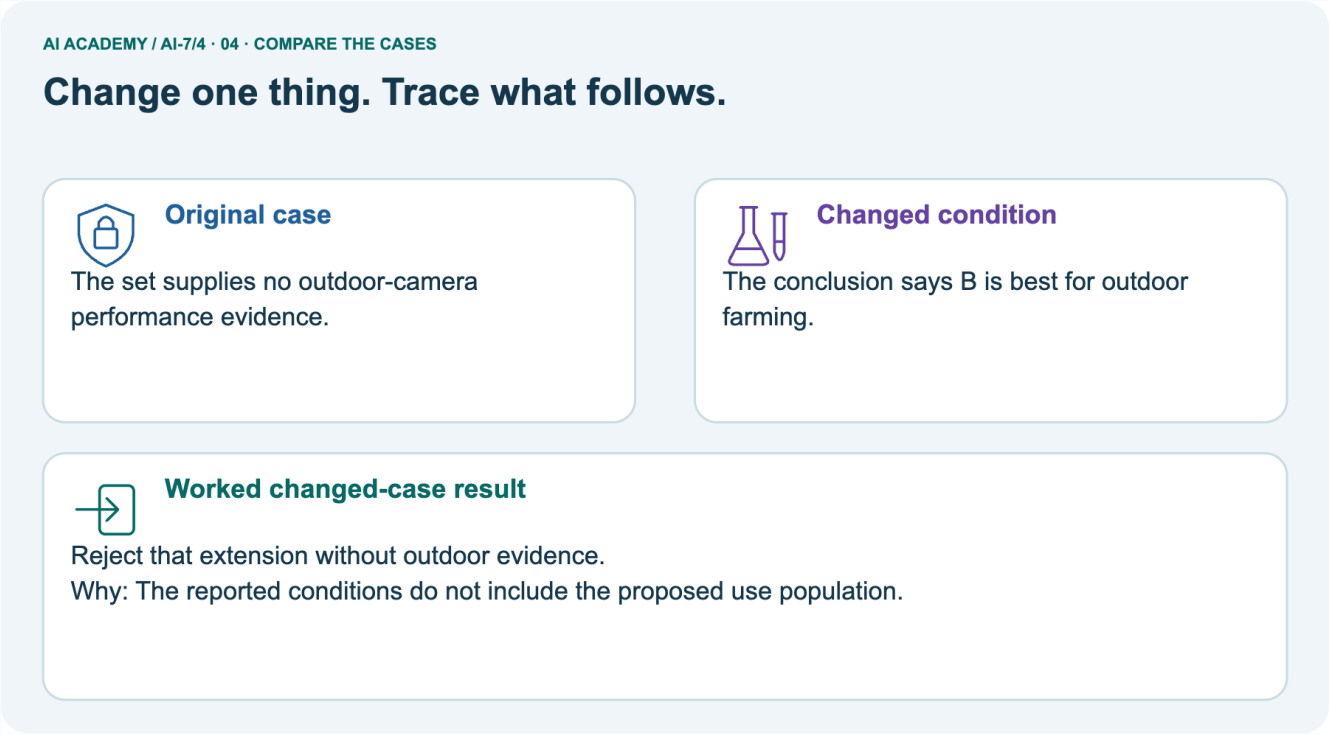

Follow the labelled picture with the matching explanation. Open the vector diagram

Two models on five shared photos. Units: correct photos out of 5. A: 3. B: 4.

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

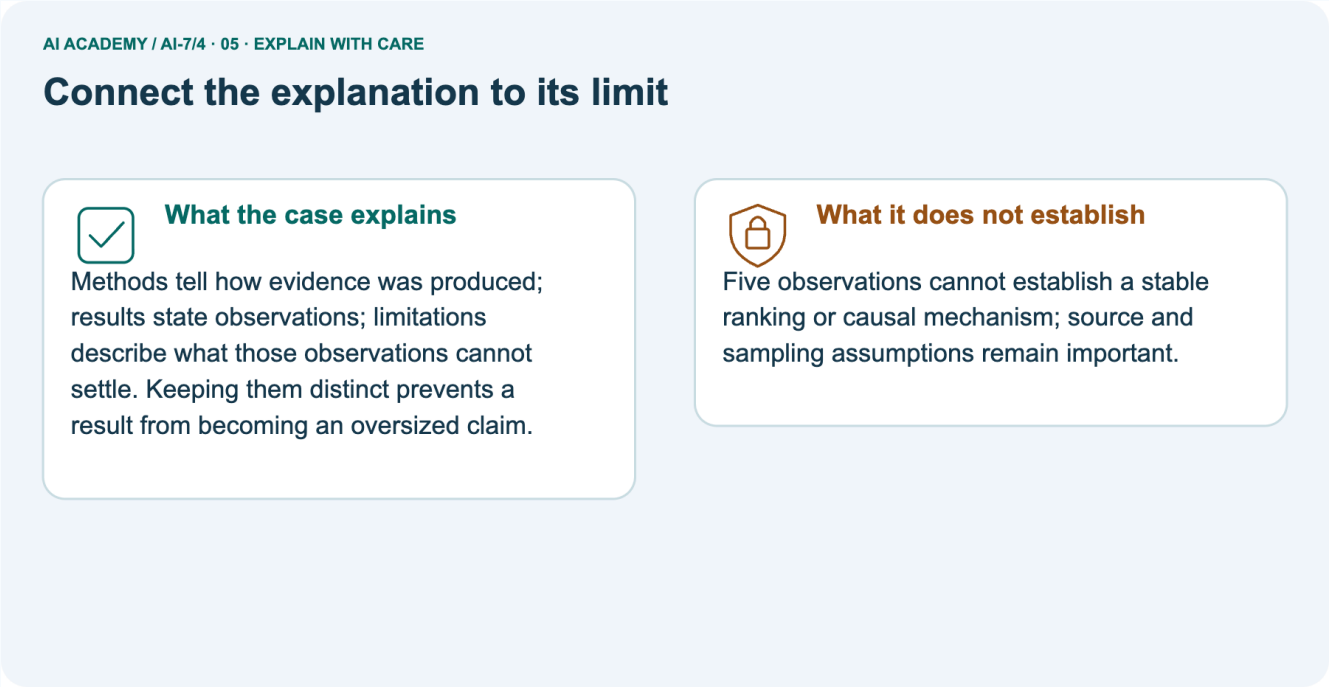
Original case: The set supplies no outdoor-camera performance **evidence**. Changed condition: The conclusion says B is best for outdoor farming. Expected **result**: Reject that extension without outdoor **evidence**.

Reason: The reported conditions do not include the proposed use population.

Predict before checking the worked changed-case **result**. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

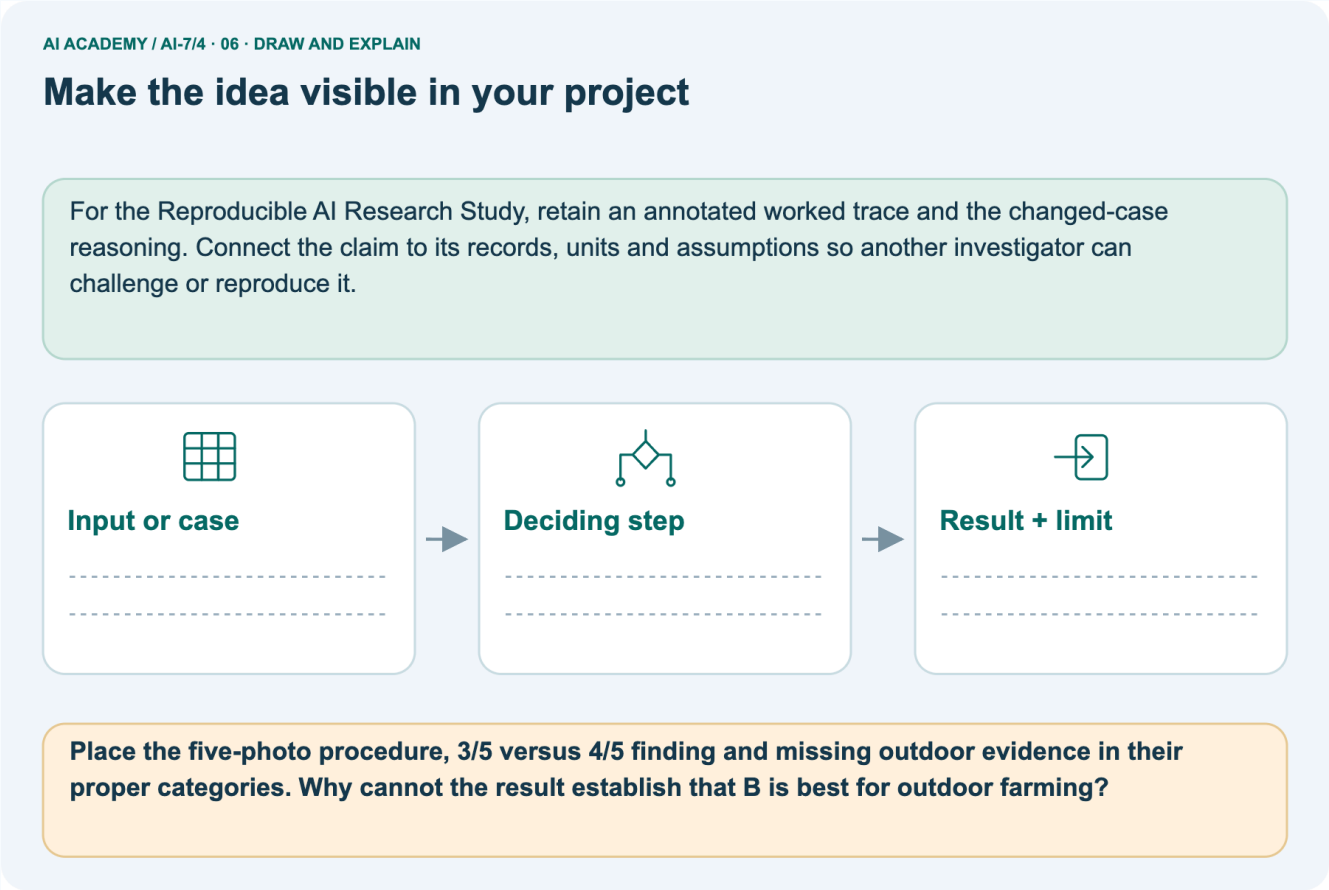
**Method**s tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

Limit: Five **observation**s cannot establish a stable ranking or causal **mechanism**; source and sampling assumptions remain important.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, **result** and limit. Supported question: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories.

Why cannot the **result** establish that B is best for outdoor farming?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

**01. Methods results and limitations: the case**

**02. Follow the deciding steps**

**03. Two models on five shared photos**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.